

DHCP Lab

1) Title: DHCP Server Configuration using Router in Cisco Packet Tracer

2) Aim / Objectives :

To configure and test Dynamic Host Configuration Protocol (DHCP) using a router in Cisco Packet Tracer for automatic IP address assignment.

3) Devices Used :

- 1 Router (2911)
- 1 Switch (2960)
- 3 PCs
- Copper Straight-through Cables

4) Topology :

PC0

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PC1 ---- Switch ---- Router

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PC2

5) IP Addressing Table :

Device	IP Address
Router G0/0	192.168.1.1

PC0	Assigned Automatically
PC1	Assigned Automatically
PC2	Assigned Automatically

Subnet Mask:- 255.255.255.0

DNS Server:- 8.8.8.8

6) Procedure / Steps :

- Added 3 PCs, one switch, and one router in Cisco Packet Tracer.
- Connected devices using copper straight-through cables.
- Configured router interface. Open router CLI and enter:-
enable
configure terminal

```
interface gigabitEthernet0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
```

- Configure DHCP Service. Configure DHCP pool:-
ip dhcp excluded-address 192.168.1.1 192.168.1.20

```
ip dhcp pool LABPOOL
network 192.168.1.0 255.255.255.0
default-router 192.168.1.1
dns-server 8.8.8.8
```

```
end
```

- Configure PCs:-
For each PC:

Desktop

IP Configuration

Select DHCP

The PCs will automatically receive IP addresses.

- Verify DHCP Assignment
- Test connectivity. From PC0:- ping 192.168.1.22 OR 192.168.1.23. Successful replies indicate proper DHCP configuration.

7) Result :

DHCP was successfully configured on the router. All client PCs automatically received IP addresses, subnet mask, default gateway, and DNS server information.

8) Concepts Learned :

- DHCP Configuration using Router
- Dynamic IP Address Allocation
- DHCP Pool Creation
- Automatic Network Configuration
- Client-Server Communication
- Network Connectivity Testing

9)Screenshot :

